	Raw Data (BC)	Time-Trend Residualized (BC)	Residualized against Functional Null (BC)	Reactivity (BC)
Min.	-4.234	-2.476	-3.323	-2.650
1st Qu.	0.105	-0.041	0.002	-0.015
Median	0.187	0.073	0.110	0.111
Mean	0.352	0.114	0.183	0.313
3rd Qu.	0.406	0.168	0.269	0.352
Max.	4.881	4.881	3.886	4.869
Wilcoxon p (H1: > 0)	0.000	0.000	0.000	0.000

Table A1: Robustness tests using alternative version of the practical emergence criterion.

using $K = 3$ bins for our main analyses. We find consistent results and support for the presence of emergence using three bins (instead of two as in the main analyses).

Emergence Capacity. 20% of individual experiment p-values are below 0.05 and Fisher test is highly significant (the largest p value is in the **ToM** condition with 4.47×10^{-3} the others are much smaller). Wilcoxon signed rank test for $\mu > 0$ of the bias-corrected and time-trend demeand data are as follows. Overall: $p = 0.005$; **Plain**: $p = 0.0004$; **Persona**: $p = 3.2 \times 10^{-6}$; **ToM**: $p = 0.991$. Wilcoxon signed rank test for $\mu > 0$ of the bias-corrected and functional-null demeand data are as follows. Overall: $p = 0.948$; **Plain**: $p = 0.999$; **Persona**: $p = 0.896$; **ToM**: $p = 0.025$. Using this stricter null indicates that only the **ToM** condition exhibits complementary cross-agent structure not explained by shared trends or independent feedback-following. That is, the stricter test isolates the construct of interest—synergy beyond coordination-free dynamics—and **ToM** is the only condition that passes it. The high p-value for **ToM** under the simpler time-trend demeaning should be taken as a methodological sensitivity (sparsity and a fat null). Together they show the **ToM** effect is robust to deconfounding but fragile under known low-power discretization. This is also in line with the stricter early synergy tests presented in Section A.12 which are also only passed by the **ToM** condition.

Practical Criterion. 5.8% of individual experiment p-values are below 0.05 and Fisher test is highly significant ($p < 10^{-16}$). Wilcoxon signed rank test for $\mu > 0$ of the bias-corrected and time-trend demeand data are as follows. Overall: $p = 3.6 \times 10^{-8}$; **Plain**: $p = 3.1 \times 10^{-5}$; **Persona**: $p = 0.001$; **ToM**: $p = 0.009$.

A.9 Additional Results on Dynamical Mechanism

The main paper shows results for Total Stability: I_3 normalized by the entropy of the macro signal. Here we show results for I_3 as well (Figure A2). They are nearly identical given that the entropy of macro signal if ≈ 1 given quantile binning with two bins.

A.10 Are Identities Persistent?

If some agents learn faster than others, even without persistent "roles," synergy can emerge simply because differentiated trajectories create complementarity across time. As indeed the mixed model test shows that 32% of groups have significant slopes provides strong evidence for the presence of heterogeneous learning rates. That is, heterogeneous learning rates could be a confound explaining some of our observed synergy effects. To rule out this alternative explanation we explore if agents have *persistent* "identities." We find the block-shuffle nulls show more significant G_3 than the full row shuffle. This suggests that synergy depends on temporal continuity of identities (not just on different learning rates at each timepoint). Furthermore, the hierarchical model test shows significant variance even without varying slopes, indicating stable role-like differences beyond learning rates. Finally, our analyses removing time trends and the functional null-model also remove learning rate slopes (Table A1). The functional null-model in particular does this in a non-linear way that is directly tied to our task. In summary, we find both heterogeneous learning rates and stable identity-linked differentiation. Heterogeneous learning rates alone cannot explain the observed emergent synergy that persist across time (providing additional insights regarding **RQ2**).

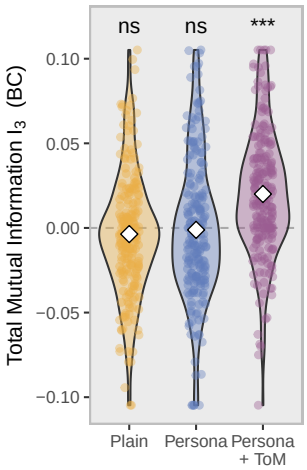

Figure A2: I_3 mutual information (time-trend demean, bias corrected) with Wilcoxon > 0 p-value indicators.

Together, the results suggest a nuanced picture of emergence in multi-agent systems. **Persona** and **ToM** produce agents with distinct, stable identities—moving agents on the differentiated-undifferentiated axis in Figure 1a. This differentiation shapes the space in which other agents can adapt, and specialization by one agent constrains the (useful) actions of the others (agents mutually constrain each other; cf. Tollefsen et al., 2013)—shifts on the interdependent-complementary axis. When combined with **ToM**, mutual adaptation both strengthens agent differentiation (creating the foundation for possible synergistic complementarity) while also increasing mutual alignment on shared goals (the target signal)—shifts on the integration axis. Differentiation enables synergy, but without redundancy and integration, synergy alone does not translate into higher collective performance. Agents in the **ToM** condition develop specialized, complementary roles while achieving stronger shared goal alignment: they are simultaneously more differentiated yet also more integrated along highly overlapping task-relevant information which the capacity for synergy can now exploit in productive ways.

A.11 Additional Details for Performance Analysis

What does synergy enable? In particular, does synergy affect performance? Connecting emergent properties of synergy and redundancy to performance is a challenging task for two main reasons. First, redundancy and synergy sit at opposite ends of the informational spectrum of correlations. Systems constrained by limited resources (such as energy or bandwidth) cannot maximize both simultaneously. The "informational budget" of correlations can be spent on shared overlap (redundancy) or on higher-order joint structures (synergy), but not both simultaneously (Williams & Beer, 2010; Varley et al., 2023; Luppi et al., 2024). Second, emergent properties are dichromatic—they require time to evolve (Bedau, 2002)—making them endogenous with run-length of the simulation. Namely, simulations in which the multi-agent system performs worse and takes longer to solve the task (or fails to solve the task altogether) provide them with more time to evolve sophisticated group coordination patterns; while successful runs terminate early, giving collectives less time to evolve emergent properties. This complicates meaningful analyses as it can make it appear that higher synergy is correlated with failure. In technical terms: time is an endogenous variable and confounds both performance and observed values of synergy and redundancy.

We address these issues as follows. First, we focus our analysis on the interaction of synergy and redundancy, rather than each variable individually to capture the tradeoff between the two. Second, we compute both emergent properties (synergy and redundancy) only on early rounds (in our case, 10 time steps). We chose 10 rounds as a good tradeoff between giving the multi-agent systems enough time to evolve, while limiting the amount of runs that have already completed. While this reduces our ability to detect the full emergent potential of the multi-agent system (since agents have less time to evolve) it creates a clean apples-to-apples comparison between measured values (they are all

based on exactly 10 time steps). In that sense, our analyses likely underestimate the full effect of synergy because we only capture the early onset of synergy. Focusing on only the first 10 time steps, leads to additional complications, however. About 23% of experiments finish early and do not make it till round ten (roughly balanced across treatments with 22% in **Plain**, 25% in **Persona**, and 20% in **ToM**). Simply removing these observations would bias our results since this censoring occurs post-treatment assignment: removing those observations would introduce sampling bias. Instead, we take these observations as censored (i.e., we cannot compute the early synergy measure) and control for this censoring using stabilized inverse probability weighting (IPW; Wooldridge, 2002). That is, in a separate modeling step we predict (via logistic regression and only task difficulty as exogenous predictor variables that is independent of the treatment) the likelihood of observing (early) synergy and redundancy.[5] We then take the inverse of the predicted likelihood as weight in the causal mediation analysis. This adjust observations in the primary analysis with the inverse of the probability of making such observations.

Our first analysis is a plain quasibinomial regression on the binary success vs. failure outcome[6] with *synergy*, *redundancy*, their interaction, task difficulty as controls, and dummy indicators for the three treatment conditions. We use our most conservative estimate of synergy and redundancy with the bias-corrected estimates with high null model sampling ($B = 2,000$), and the conservative MMI redundancy (Mediano et al., 2025) with MI estimated with Miller–Madow bias correction (Miller, 1955). Results are robust to using MI redundancy and Jeffreys-smoothed probabilities.

For our second analysis we perform causal mediation analysis (Imai et al., 2010) to link our interventions to performance, via multi-agent synergy as a mechanism. We estimate the model using the `{mediation}` package in **R** (Tingley et al., 2014). We find that the **ToM** intervention has a significant indirect effect on performance through its impact on synergy, beyond any direct treatment effects.

A.12 Robustness Tests using "Early Synergy"

Since our experiment is censored when groups correctly guess the target number, length of data in our analyses varies between groups. As robustness test, we also compute "early synergy" using only data up to round 15 (removing about 40% of the data). Using the practical criterion, we find about 5% of individual p-values are below 0.05. Fisher test is highly significant overall ($p = 2.7 \times 10^{-6}$), significant in the **Plain** (0.001) and **ToM** condition ($p = 0.0004$) and marginally significant in the **Persona** condition ($p = 0.059$). Using a more aggressive cutoff in round 10 (dropping only about 20% of trials but giving multi-agent systems less time to evolve synergy and having less data for estimation we find only significant in the **ToM** condition ($p = 0.028$).

A.13 Experiments with other Models

To test generalizability of our findings, we conducted experiments using four other models:

1. Llama 3.1 8B,
2. Llama 3.1 70B Instruct,
3. Google Gemini 2.0 flash, and
4. Qwen3 235B A22B Instruct 2507.

We conducted 100 replications of the group experiment with ten agents and temperature of 1.0 of each model and each treatment condition.

Before turning to comparative results, we first address a structural challenge that emerged with Qwen3—one that speaks to a broader issue with deploying reasoning models in multi-agent coordination settings.

[5] As measures for task difficulty we use (a) how far groups target number was from the mid point (smaller numbers requiring more steps of the binary search process); and (b) the modulo remainder of the target number and the group size (target numbers that are closer to divisibility by the group size are easier to guess by pure chance).

[6] While the raw outcome is binary and binomial regression would be appropriate, after the IPW weighting step, outcomes are no longer binary, hence quasibinomial model.